MULTI-CROP DATABASE SYSTEM FOR VARIETY RECOMMENDATION IN SUB-SAHARAN AFRICA

DATA MANAGEMENT SYSTEMS PROJECT

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1. Problem Identification

Overview

The Multi-Crop Database System (MDS) is a relational database and analytics platform designed to support evidence-based crop variety selection across Sub-Saharan Africa (SSA). The system integrates multi-year, multi-location trial data from five countries for four major crops; to help researchers, breeders, agronomists, and farmers make data-driven decisions.

Goal of This Project

The goal of this project is to design and implement a relational database management system that organizes and integrates multi-crop breeding and MET data for maize, rice, sorghum, and cowpea across five SSA countries (Nigeria, Kenya, Ethiopia, Tanzania, and Ghana).

Specific Objectives

The specific objectives were to:

i. Create a centralized, structured database for storing multi-environment trial data, environmental variables, and crop variety information.
ii. Standardize heterogeneous datasets from multiple sources into a consistent relational format.
iii. Enable efficient querying to answer breeding, agronomic, and decision-making questions.
iv. Support downstream machine learning pipelines for predicting the best variety for a given location and growing season.
v. Enable future integration into a decision-support mobile app for farmers.

Key Decision-Support Questions the Database Can Answer

The database enables users to perform complex SQL queries to support breeding, agronomy, and farmer-oriented decision-making. Examples of core questions include:

i. Which varieties perform best across diverse environments?
ii. Which varieties show the greatest stability or adaptability across locations and years?
iii. How do varieties perform under specific stresses such as drought, low nitrogen, or optimum conditions?
iv. What is the expected yield of a variety in a specific season (e.g., dry vs. rainy)?
v. How do traits such as days to maturity, plant height, anthesis–silking interval (ASI), and stay-green vary across environments?

vi. Which breeding institutions consistently produce the best-performing varieties?

vii. Which locations are most suitable for cultivating specific varieties?

viii. What are the year-to-year trends in major agronomic traits?

ix. How can breeders and farmers use historical MET data for evidence-based decisions?

2. Data Identification and Collection

Data Description

The dataset used for constructing the Multi-Crop Database System consists of multi-environment trial (MET) data collected across five Sub-Saharan African countries: Nigeria, Kenya, Ethiopia, Tanzania, and Ghana, for four major crops: maize, rice, sorghum, and cowpea. The data represent multi-year, multi-location field trials conducted under diverse environmental conditions, with accompanying environmental and agronomic metadata including trial metadata, phenotypic measurements, and soil characteristics

The original datasets were obtained from collaborating institutions, including the University of Ilorin (Nigeria), the Institute for Agricultural Research, Ahmadu Bello University, Zaria (Nigeria), the International Institute of Tropical Agriculture (IITA), and national research stations across Kenya, Tanzania, Ghana, and Ethiopia. Environmental data were obtained from the official environmental agencies of each respective country for each trial year. For locations or traits where data were incomplete, missing, or inconsistent across sites, simulated values were generated to ensure completeness and maintain structural consistency.

Data Volume Summary

1. **Maize:** Data for maize comes from MET of 237 varieties across four maturity groups: early, extra-early, intermediate, and late. Trials were conducted across five locations in Nigeria (Ibadan, Ilorin, Mokwa, Zaria, and Omu-Aran) and three locations each in Kenya (Kiboko, Kitale, and Nairobi), Ethiopia (Bako, Hawassa, and Melkassa), Ghana (Damongo, Nyankpale, and Yendi), and Tanzania (Arusha, Morogoro, and Dodoma) from 2020 to 2022. Trials were conducted under three key environmental conditions: Low-Nitrogen, Drought, and Optimum.
2. **Cowpea:** The cowpea dataset comprises MET data of 50 varieties evaluated across three locations in Ghana and five locations in Nigeria (same locations as for maize) in 2020, 2021, and 2022. All cowpea trials were conducted exclusively under Optimum environmental conditions.

3. **Rice:** The dataset for rice consists of data obtained from MET of 60 varieties evaluated across multiple regions in three countries: Kenya (Kiboko, Kitale, and Nairobi), Ghana (Damongo, Nyankpale, and Yendi), and Nigeria (Ibadan, Ilorin, Mokwa, Zaria, and Omu-Aran). All rice trials were conducted exclusively under Optimum environmental conditions. The dataset covers a three-year period from 2020 to 2022.
4. **Sorghum:** The sorghum dataset comprises 60 varieties evaluated across four countries. Trials were conducted in Ghana (Damongo, Nyankpale, Yendi), Kenya (Kiboko, Kitale, Nairobi), Tanzania (Arusha, Morogoro, Dodoma), and Nigeria (Ibadan, Omu-Aran). Sorghum was tested under three environmental conditions: Low-N, Drought, and Optimum. The dataset covers a three-year period (2020–2022).

Types of Data Collected

The Multi-Crop Database System captures multiple layers of information for each trial, covering crop, location, environmental conditions, soil properties, and phenotypic traits. The primary data categories include:

a. **Crop Data:** Information about each crop variety, including variety name and entry, pedigree information, breeding institution and maturity group (where applicable)
b. **Location Data:** Metadata for each trial site, includingcountry and region, agro-acological zone, elevation (meters above sea level).
c. **Environmental Data:** Trial-specific environmental conditions, including season (e.g., rainy, dry), environmental condition (e.g., Optimum, Low-N, Drought), rainfall (mm), mean temperature (°C).
d. **Soil-Related Data:** Properties of the soil at each trial location soil type (classification), soil pH, soil nutrient content: Nitrogen (N), Phosphorus (P), Potassium (K).
e. **Phenotypic Measurement (Traits and Values):** A wide array of agronomic traits was recorded, depending on the crop. These include:
    - **Common traits across crops: g**rain yield, plant height, days to flowering (anthesis and silking for maize).
    - **Crop-specific traits:**
        - **Maize:** Anthesis-silking interval, ear height, husk cover, stay-green, grain moisture, ear aspect, plant aspect, grain moisture.
        - **Rice:** Tiller number, panicle length, panicle weight, 1000-grain weight, 100-grain weight

- **Sorghum:** Grains per panicle, panicle width, panicle number, stay-green, dry matter
- **Cowpea:** Pods per plant, peduncles per plant, seeds per plot, harvest index.

All phenotypic measurements are recorded as numeric values where possible. Traits not measured in a particular trial or specific variety are represented as NULL in the database to maintain consistency across datasets.

3. Data Cleaning and Transformation

The raw datasets underwent rigorous cleaning and transformation using both Excel and OpenRefine before integration into the database. Raw data was in multiple formats, including spreadsheets, CSV files, institutional databases and technical reports. The following steps were performed to ensure the data were consistent, standardized, and ready for relational database modeling:

- Harmonizing datasets from different countries and crops into a unified template with a consistent column structure.
- Standardizing crop names, variety identifiers, and trait labels to eliminate variations caused by spelling, abbreviations, or local naming conventions.
- Handling missing values and correcting inconsistent entries, including the treatment of outliers and invalid measurements.
- Converting measurement units (where applicable) to ensure uniformity across datasets.
- Removing duplicate or conflicting entries identified during cross-checking of multi-location and multi-year data.
- Normalizing environmental variables such as rainfall, soil type, and stress conditions for consistency.
- Creating additional placeholder columns for traits that were not measured in some crops, ensuring all crop datasets aligned to the same column structure before merging.
- Replacing empty fields with 'NA' in the preprocessing stage (later converted to NULL during database import).
- Simulating missing trial data where necessary (e.g., incomplete environmental records) to maintain structural consistency across datasets.

4. Database Design

The database schema follows a star-schema approach, with a central fact table for phenotypic measurements and related dimension tables for crops, varieties, locations, trials, and traits.

Entity–Relationship (ER) Model

The database consists of six core entities in addition to the master crop table, which served as the primary staging table from which all other dimension and fact tables were derived:

1. **Crop:** Stores information about each crop i ncluded in the system. This table allows the system to differentiate among crops like maize, rice, sorghum, and cowpea.

   Key Fields: crop_id (PK), crop_name, scientific_name
2. **Location:** Represents trial sites across countries and regions. Locations are linked to trials to capture environmental context and geographical metadata.

   Key Fields: location_id (PK), country, region, agro_ecological_zone, elevation, soil_type
3. **Variety:** Contains detailed information on individual crop varieties. Each variety is associated with a specific crop, allowing queries across or within crops.

   Key Fields: variety_id (PK), entry, name, pedigree, breeding_institution, maturity_group, crop_id (FK)
4. **Trial:** Stores metadata about experimental trials conducted in specific locations, years, seasons, and environmental conditions. Trials serve as the context for phenotypic measurements and allow multi-environment comparison.

   Key Fields: trial_id (PK), location_id (FK), crop_id (FK), year, season, environment_condition, rainfall_mm, mean_temperature, soil_pH, soil_N_content, soil_P_content, soil_K_content
5. **Trait:** Defines the phenotypic traits measured in trials. This dimension table ensures consistent naming and descriptions of traits.

   Key Fields: trait_id (PK), trait_name, trait_description, primary_source
6. **Phenotype Measurement:** Stores actual trait values collected in trials for specific varieties. This is the core fact table for analytics. It supports aggregation, filtering, and multi-trait analysis across trials, locations, and crops.

   Key Fields: measurement_id (PK), trial_id (FK), variety_id (FK), trait_id (FK), value

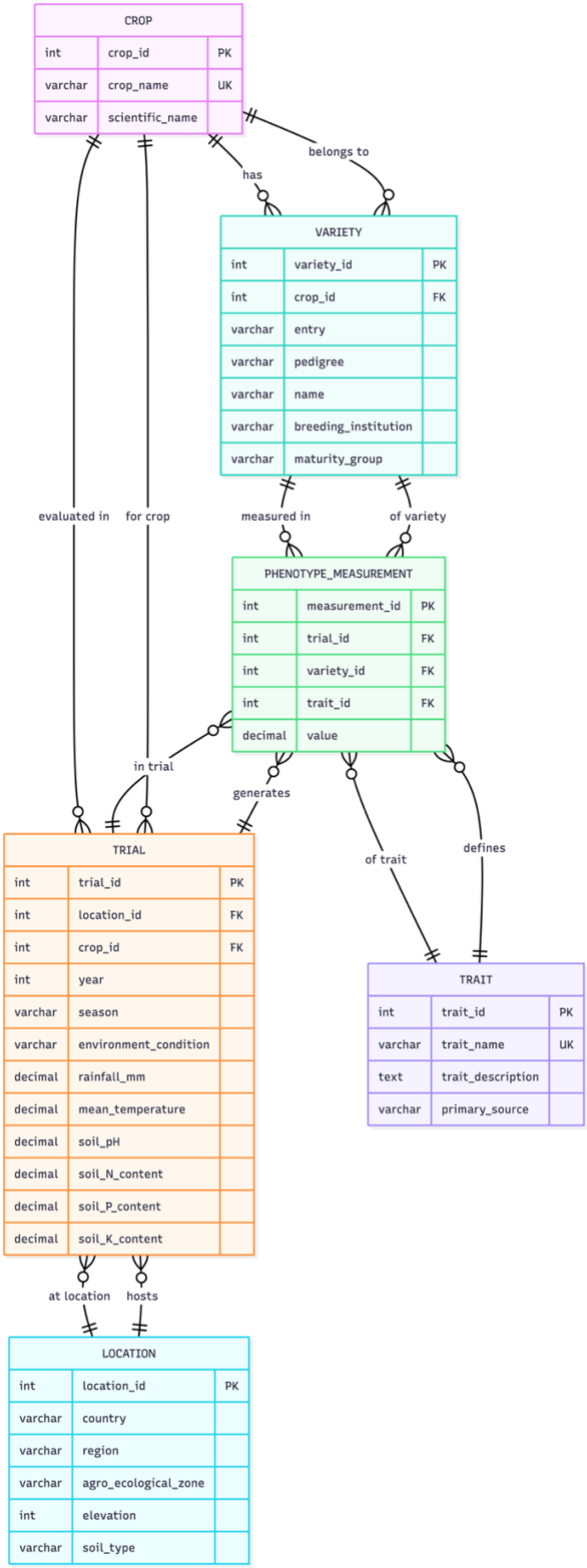

Figure 1: ER Diagram of the Multi-Crop Database System

Relationships

One-to-Many Relationships:

Crop → Variety: One-to-many. Each crop can have multiple varieties.

variety.crop_id → crop.crop_id

Crop → Trial: One-to-many. Each crop can be evaluated in multiple trials.

trial.crop_id → crop.crop_id

Location → Trial: One-to-many. Each location can host multiple trials.

trial.location_id → location.location_id

Variety → Phenotype Measurement: One-to-many. Each variety has multiple measurements across different trials.

phenotype_measurement.variety_id → variety.variety_id

Trial → Phenotype Measurement: One-to-many. Each trial generates multiple measurements across different varieties and traits.

phenotype_measurement.trial_id → trial.trial_id

Trait → Phenotype Measurement: One-to-many. Each trait is measured multiple times across different varieties and trials.

phenotype_measurement.trait_id → trait.trait_id

Many-to-Many Relationships (via Junction Tables):

Crop ↔ Location (via trial): Multiple crops can be tested at one location, and one crop can be tested at multiple locations.

Variety ↔ Trial ↔ Trait (via phenotype_measurement): The fact table implements a three-way many-to-many relationship. Each measurement uniquely links one variety, one trial, and one trait, enabling multi-dimensional analysis of variety performance across environmental conditions and phenotypic characteristics.

Referential Integrity:

All foreign key relationships enforce ON DELETE RESTRICT to prevent orphaned records and maintain data consistency across related tables.

Normalization

The database is designed following standard normalization principles:

1NF (First Normal Form): Each column contains atomic values; no repeating groups exist.

2NF (Second Normal Form): All non-key attributes are fully functionally dependent on the primary key.

3NF (Third Normal Form): No transitive dependencies exist; each attribute depends only on the primary key.

5. Database Implementation

Methods and Steps to Process Data and Implement the Database

The creation of the Multi-Crop Database System involved multiple stages. Following data collection, cleaning and transformation, outlined in sections 2 and 3, the workflow is summarized below.

Implementation of the Database

To recreate the database from raw CSV data files, the following steps should be followed:

Prerequisites

- Database Server: MariaDB installed and running
- SQL Client: DBeaver
- Data Files: All cleaned CSV files (e.g., master_crop_dataset.csv, trait_description.csv, etc.)

Step 1: Create the Database

Begin by creating a new database schema in MariaDB using the SQL script provided in the project directory.

Example:

CREATE DATABASE Crops_MET;

USE Crops_MET;

Step 2: Create and Populate the Master Crop Table

Use the scripts 01_create_master_crop_table.sql provided inside the sql_scripts directory to create the master crop table. Then populate it using the 02_load_data_into_master_crop_table.sql script inside the same directory. This table serves as the primary staging table from which all other dimension and fact tables will be derived.

Step 3: Create Remaining Tables and Populate Them

After the master table is loaded, create and populate all remaining tables using the SQL scripts available in the sql_scripts directory. These scripts are arranged in sequential order:

a. Crop Table → Script: 03_create_crop_dimension_table.sql

b. Location Table → Script: 04_create_location_table.sql

c. Variety Table → Script: 05_create_variety_table.sql

d. Trial Table → Script: 06_create_trial_table.sql

e. Trait Table → Script: 07_create_trait_table.sql

f. Index formation → Script: 08_add_indexes_script.sql

g. Phenotype Measurement Table → Script: 09_create_phenotype_measurement_table.sql

Step 4: Verify Data

Run sample SELECT queries on each table to ensure data was loaded correctly.

Step 5: Backup the Database

A final database dump (database_dump.sql) was created to preserve the complete state of the database for sharing and query execution.

6. Data Use

Example Outputs of Data use Queries

Query 1: List all varieties for a selected crop (Maize)

```
SELECT v.variety_id, v.name, v.maturity_group, c.crop_name
FROM variety v
JOIN crop c ON v.crop_id = c.crop_id
WHERE c.crop_name = 'Maize'
LIMIT 5;
```

OUTPUT (Head)

variety_id	name	maturity_group	crop_name
51	SAMMAZ-01	Extra-Early	Maize
52	SAMMAZ-02	Extra-Early	Maize
53	SAMMAZ-03	Extra-Early	Maize
54	SAMMAZ-04	Early	Maize
55	SAMMAZ-05	Early	Maize

Query 2: Average grain yield by region

```
SELECT l.region, AVG(p.value) AS avg_yield
FROM phenotype_measurement p
JOIN trait t ON p.trait_id = t.trait_id
JOIN trial tr ON p.trial_id = tr.trial_id
JOIN location l ON tr.location_id = l.location_id
WHERE t.trait_name = 'grain_yield'
GROUP BY l.region
LIMIT 5;
```

Output (Head)

region	avg_yield
Arusha	1362.728039
Bako	1512.704102
Damongo	1210.892982
Dodoma	1348.163093
Hawassa	1513.322413

Query 3: Top-performing maize variety under drought stress

```
SELECT v.name AS variety, tr.environment_condition, p.value AS yield
FROM phenotype_measurement p
JOIN trial tr ON p.trial_id = tr.trial_id
JOIN variety v ON p.variety_id = v.variety_id
JOIN trait t ON p.trait_id = t.trait_id
WHERE tr.environment_condition = 'Drought'
  AND t.trait_name = 'grain_yield'
ORDER BY p.value DESC
LIMIT 5;
```

Output (Head)

variety	environment_condition	yield
ILOMAZ-57	Drought	16305.88
TELA-41	Drought	16305.88
TVu-1259	Drought	15952.59
SAMMAZ-41	Drought	15952.59
Komboka	Drought	15952.59

Query 4: Top-performing cowpea variety across all regions

```
SELECT v.name AS variety_name, v.breeding_institution, l.country, l.region,
AVG(pm.value) AS avg_grain_yield
FROM phenotype_measurement pm
JOIN trait tr  ON pm.trait_id = t.trait_id
JOIN variety v ON pm.variety_id = v.variety_id
JOIN trial t ON pm.trial_id = t.trial_id
JOIN location l  ON t.location_id = l.location_id
WHERE tr.trait_name = 'grain_yield' AND c.crop_name = 'Cowpea'
GROUP BY  v.variety_id, v.name, v.breeding_institution, l.country, l.region
ORDER BY avg_grain_yield DESC
LIMIT 5;
```

Output (Head)

variety_name	breeding_institution	country	region	avg_grain_yield
TVu-1192	IAR	Ethiopia Bako		2505.378889
TVu-14261	WACCI	Nigeria	Zaria	2411.538667
IT86D-1010-1	IITA	Nigeria	Zaria	2381.880667
IT82D-889	IITA	Nigeria	Zaria	2282.676667
TVu-14677	WACCI	Ethiopia Melkassa		2276.157222

Query 5: Year-to-Year Yield Trend for a Crop

```
SELECT t.year, c.crop_name, AVG(pm.value) AS avg_yield
FROM phenotype_measurement pm
JOIN trait tr  ON pm.trait_id = t.trait_id
JOIN variety v ON pm.variety_id = v.variety_id
JOIN crop c ON v.crop_id = c.crop_id
JOIN trial tr ON pm.trial_id = tr.trial_id
JOIN location l ON t.location_id = l.location_id
 -- Core Filter
WHERE tr.trait_name = 'grain_yield' AND c.crop_name = 'Maize'
 -- Optional Filter
  AND l.country = 'Nigeria'
  AND tr.environment_condition = 'Optimum'
GROUP BY  t.year, c.crop_name
ORDER BY t.year ASC;
```

OUTPUT (Head)

year	crop_name	avg_yield
2020	Maize	1908.677464
2021	Maize	1566.359057
2022	Maize	1682.427191